Lecture 24 — Modular Congruence

CS 173: Discrete Structures · Number Theory

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1 Definition of Congruence Modulo n

Definition 1.1 (Congruence modulo n)**.** Given $x, y \in \mathbb{Z}$ and $n \in \mathbb{N}^+$, we say x is *congruent to y modulo n*, written

$$x \equiv y \pmod{n},$$

if and only if $n \mid (x - y)$.

Remark. The symbol $\equiv$ here denotes an *equivalence relation* on $\mathbb{Z}$. Think of it as a structured version of $=$: rather than asking whether two integers are identical, we ask whether they leave the same remainder when divided by n. The biconditional $p \equiv q$ uses $=$ as the underlying sign of equality at the level of equivalence classes.

2 Congruence is an Equivalence Relation

Theorem 2.1 (Reflexivity)**.**

$$\forall x \in \mathbb{Z}, \quad x \equiv x \pmod{n}.$$

$x - x = 0$ and $n \mid 0$ for all $n \in \mathbb{N}^+$. □

Theorem 2.2 (Symmetry)**.**

$$\forall x, y \in \mathbb{Z}, \quad x \equiv y \pmod{n} \Rightarrow y \equiv x \pmod{n}.$$

If $n \mid (x - y)$, then $n \mid (-(x - y)) = (y - x)$. □

Theorem 2.3 (Transitivity)**.**

$$\forall x, y, z \in \mathbb{Z}, \quad \big(x \equiv y \pmod{n} \wedge y \equiv z \pmod{n}\big) \Rightarrow x \equiv z \pmod{n}.$$

$n \mid (x - y)$ and $n \mid (y - z)$ imply $n \mid \big((x - y) + (y - z)\big) = x - z$. □

3 Congruence and Addition

Theorem 3.1 (Addition compatibility)**.**

$$(\forall n \in \mathbb{N}^+)\ (\forall x, y \in \mathbb{Z})\ (\forall z \in \mathbb{Z}) \qquad x \equiv y \pmod{n} \ \Rightarrow\ x + z \equiv y + z \pmod{n}.$$

Theorem 3.2 (Additive cancellation)**.** Addition cancellation holds unconditionally:

$$x + z \equiv y + z \pmod{n} \ \Rightarrow\ x \equiv y \pmod{n}.$$

4 Congruence and Multiplication

Theorem 4.1 (Multiplication compatibility)**.**

$$(\forall n \in \mathbb{N}^+)\ (\forall x, y \in \mathbb{Z})\ (\forall z \in \mathbb{Z}) \qquad x \equiv y \pmod{n} \ \Rightarrow\ x \cdot z \equiv y \cdot z \pmod{n}.$$

Caution (Multiplicative cancellation is *not* always valid)**.** The converse requires an extra hypothesis. In general,

$$x \cdot z \equiv y \cdot z \pmod{n} \ \text{and}\ z \not\equiv 0 \pmod{n} \ \not\Rightarrow\ x \equiv y \pmod{n}.$$

Cancellation *is* valid when $\gcd(z, n) = 1$:

$$x \cdot z \equiv y \cdot z \pmod{n} \ \text{and}\ \gcd(z, n) = 1 \ \Rightarrow\ x \equiv y \pmod{n}.$$

5 The Set $\mathbb{Z}_n$ and Modular Arithmetic

5.1 Definition of $\mathbb{Z}_n$

Definition 5.1 (Integers modulo n)**.** For $n \in \mathbb{N}^+$, define

$$\mathbb{Z}_n \ =\ \{0, 1, 2, \ldots, n-1\}.$$

This is the set of *residues* modulo n. Every integer $x \in \mathbb{Z}$ is congruent to exactly one element of $\mathbb{Z}_n$.

5.2 Modular Addition and Multiplication on $\mathbb{Z}_n$

Definition 5.2 (Operations on $\mathbb{Z}_n$)**.** For $a, b \in \mathbb{Z}_n$, define:

$$a \oplus b \ =\ (a + b) \bmod n,$$
$$a \otimes b \ =\ (a \cdot b) \bmod n.$$

Results are always reduced to lie in $\{0, 1, \ldots, n-1\}$.

Example. Working in $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$:

$$\begin{aligned}
1 \cdot 2 &= 2, \\
2 \cdot 2 &= 4, \\
4 \cdot 4 &\equiv -2 \equiv 4 \pmod 6, \\
2 + 1 &= 3, \\
2 + 4 &\equiv 0 \pmod 6, \\
3 + 3 &\equiv 0 \pmod 6, \\
1 + 5 &\equiv 0 \pmod 6, \\
2 + 5 &\equiv 1 \pmod 6, \\
3 + 0 &\equiv 3 \pmod 6.
\end{aligned}$$

5.3 Algebraic Properties of $(\mathbb{Z}_n, +, \cdot)$

$\mathbb{Z}_n$ equipped with modular addition and multiplication satisfies the following properties.

Theorem 5.1 (Commutativity). For all $a, b \in \mathbb{Z}_n$:

$$a \oplus b = b \oplus a, \qquad a \otimes b = b \otimes a.$$

Theorem 5.2 (Associativity). For all $a, b, c \in \mathbb{Z}_n$:

$$(a \oplus b) \oplus c = a \oplus (b \oplus c), \qquad (a \otimes b) \otimes c = a \otimes (b \otimes c).$$

Theorem 5.3 (Distributivity). For all $a, b, c \in \mathbb{Z}_n$:

$$a \otimes (b \oplus c) = (a \otimes b) \oplus (a \otimes c).$$

Theorem 5.4 (Additive identity). The element $0 \in \mathbb{Z}_n$ is the additive identity:

$$\forall a \in \mathbb{Z}_n, \quad a \oplus 0 = a.$$

Theorem 5.5 (Additive inverse). Every element of $\mathbb{Z}_n$ has an additive inverse. For $a \in \mathbb{Z}_n$,

$$-a \;\equiv\; n - a \pmod n.$$

That is, $a \oplus (n - a) \equiv 0 \pmod n$.

Remark. The presence of additive inverses is precisely what the board examples illustrate:

$$1 + 5 \equiv 0, \quad 2 + 4 \equiv 0, \quad 3 + 3 \equiv 0 \pmod 6.$$

Each pair $(a,\, 6 - a)$ sums to 0 in $\mathbb{Z}_6$.

Theorem 5.6 (Multiplicative identity)**.** The element $1 \in \mathbb{Z}_n$ is the multiplicative identity:

$$\forall a \in \mathbb{Z}_n, \quad a \otimes 1 = a.$$

Caution (Multiplicative inverses need not exist)**.** Unlike additive inverses, a multiplicative inverse of a in $\mathbb{Z}_n$ exists if and only if $\gcd(a, n) = 1$. When n is prime, every nonzero element has a multiplicative inverse and $\mathbb{Z}_n$ forms a *field*.

6 Summary

Proof. 1. $x \equiv y \pmod n \iff n \mid (x - y)$.

2. $\equiv \pmod n$ is an equivalence relation (reflexive, symmetric, transitive).

3. Addition is fully compatible: cancellation holds in both directions.

4. Multiplication is compatible in one direction. Cancellation requires $\gcd(z, n) = 1$.

5. $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ is closed under modular $+$ and $\cdot$.

6. $(\mathbb{Z}_n, +)$ is a commutative group: it is associative, commutative, has an identity (0), and every element has an additive inverse $(n - a)$.

7. $(\mathbb{Z}_n, +, \cdot)$ satisfies distributivity, making it a commutative ring.

Remark (Looking ahead)**.** These ring-theoretic properties underpin modular arithmetic. We will use them to solve linear congruences $ax \equiv b \pmod n$ via the extended Euclidean algorithm, and to characterise when solutions exist and how many there are.